

Lab Experiment 8

Depth First Search (DFS)

Aim

To write a Python program to implement Depth First Search (DFS) traversal in a graph.

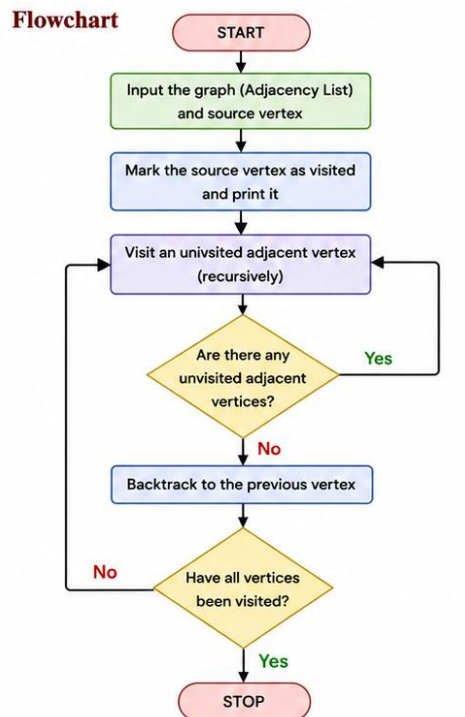
Objective

The objective of this experiment is to implement the Depth First Search algorithm for graph traversal. It aims to visit all vertices by exploring each branch as deeply as possible before backtracking.

Algorithm

1. Create a graph using an adjacency list.
2. Select the source vertex and mark it as visited.
3. Display the current vertex.
4. Visit each unvisited adjacent vertex recursively.
5. Backtrack when no unvisited adjacent vertices are available.
6. Repeat until all vertices are visited.

Flowchart



Code

```
graph = {  
    'A': ['B', 'C'],  
    'B': ['D', 'E'],  
    'C': ['F'],  
    'D': [],  
    'E': ['F'],  
    'F': []  
}
```

```
visited = set()
```

```
def dfs(vertex):
```

```

if vertex not in visited:

    print(vertex, end=" ")

    visited.add(vertex)

    for neighbor in graph[vertex]:

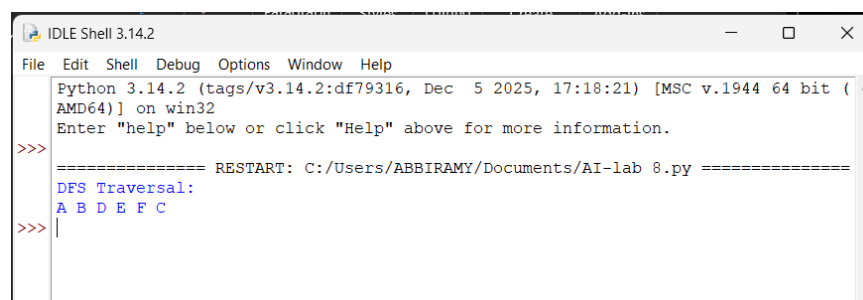
        dfs(neighbor)

print("DFS Traversal:")

dfs('A')

```

Output



The screenshot shows an IDLE Shell window titled 'IDLE Shell 3.14.2'. The window contains the following text:

```

Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/ABBIRAMY/Documents/AI-lab 8.py =====
DFS Traversal:
A B D E F C
>>>

```

Result

The Python program successfully implemented the Depth First Search (DFS) algorithm and traversed all the vertices of the graph.

Conclusion

Depth First Search (DFS) efficiently traverses a graph by exploring one branch completely before backtracking. It is widely used in graph traversal, path finding, and cycle detection.